

GABRIEL D. WEST

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Computer architecture, microarchitecture, hardware-software co-design, RTL, FPGA synthesis, and systems tooling.

EDUCATION

University of Illinois Urbana-Champaign

Expected May 2029

B.S. Computer Engineering

Overall GPA: 4.0

National Merit Finalist · AP Scholar with Distinction

TECHNICAL SKILLS

Languages: C++, C, Verilog, SystemVerilog, Python, Assembly, Java

Hardware / EDA: Vivado, Logisim, Nexys A7 FPGA, GTKWave

Tools: Git, CMake

INDEPENDENT PROJECTS (portfolio: gabrielwest.dev)

fpga-builder: Graphical node editor with synthesizable Verilog generator

May 2026 – July 2026

C++ / ImGui / Verilog / SystemVerilog / Vivado | github.com/gw12343/fpga-builder

- Built a C++/ImGui digital-logic node editor with hierarchical modules and live RTL preview.
- Developed graph-to-Verilog compiler with const-fold, dead-code strip, and combinational-loop rejection.
- Validated by synthesizing VGA controller, UART, and 32-bit CPU in Vivado and running on FPGA, zero manual edits.
- Verified cycle-exact agreement between golden model and unmodified CPU RTL for Fibonacci and IEEE-754 soft-float.

32-bit CPU: custom ISA, assembler, emulator, and FPGA bring-up

May 2024 – Jan 2025

C / Java / Logisim / Vivado / FPGA | github.com/gw12343/32-bit-cpu

- Designed a from-scratch 32-bit ISA and microcoded CPU (8 GPRs plus PC/SP/BP/AR/IR, 32-bit ALU, 5 flags, 4K×29-bit control ROM, interrupts, memory-mapped I/O).
- Built cycle-accurate C emulator (Nuklear/SDL2 GUI) as the golden model, loading identical program and microcode ROMs.
- Built a Java two-pass assembler (label resolution/operand type-checking/instruction validation) which specifies the ISA.
- Generated program/microcode ROMs are shared across emulator, Logisim prototype, and FPGA to stay in sync.
- Ported to FPGA via fpga-builder (2026): closed timing at 50 MHz on Nexys A7-100T (16 logic levels, +0.09 ns slack).
- Ran Snake over UART on the physical board as a live hardware demo.

cpp-engine: Modular 3D OpenGL game engine

May 2025 – Jan 2026

C++ / OpenGL / Lua | github.com/gw12343/cpp-engine

- Built a C++17 3D engine with an ECS, GUID asset pipeline, and hot reload.
- Implemented deferred rendering (shadows, SSAO, bloom, HDR IBL): ~120 FPS on a 1.8M-tri scene at 4K, RTX 3060.
- Integrated Jolt physics (rigid bodies, collision, character controller) and ozz skeletal animation.
- Designed a docked ImGui editor with Lua-driven gameplay scripting and scene serialization.

Autonomous differential-drive robot

July 2021 – Sept 2025

Python / C / Linux / ROS2 Humble / Jetson Orin Nano / ESP32 / Riva / Llama 3 | gabrielwest.dev

- Built a differential-drive robot on a Jetson Orin Nano with ESP32 encoder odometry and RPLiDAR A1.
- Fused sensor data into Nav2 and SLAM to autonomously navigate to named poses with obstacle avoidance.
- Achieved a 100 Hz motor control loop and a 20 Hz Nav2 planning loop.
- Wired voice stack (Riva STT → Llama 3 → Riva TTS) mapping unconstrained speech to Nav2 goals in under 750 ms.

LEADERSHIP

Design Lead, FIRST Robotics Competition (FRC Team 20)

Sept 2025 – June 2026

- Led the mechanical design subteam for an 80+ member FRC team, syncing with electrical/programming.
- Redesigned the robot's intake, cutting weight and improving cycle time/scoring by 30%.